

Executive Summary

HR Attrition Analysis

Overview

This analysis is focused towards attrition across 1,470 employees utilizing EDA, PowerBI dashboards & DAX measures to identify which segments carry the biggest risk of turnover / highest attrition rates.

1,470 TOTAL EMPLOYEES	237 DEPARTED	16.12% ATTRITION RATE	7 yrs AVG. TENURE
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Problem

total attrition stands at **16.12%** (237 of 1,470 employees), data on which roles, working conditions or tenure stages , etc. that actually drive that number can be used to assist with retention budget allocation.

HR efforts at retention can risk getting spread across the workforce instead of targeting segments where it will have the most impact.

Details

- **Sales Representatives** have an attrition rate of 39.76%, being a true statistical outlier with a Z-score of 2.0, against the mean attrition rate of 15.7% across all roles.
- **Employees who work overtime** (30.53% attrition rate) are 2.9 times more likely to leave in comparison to employees who do not work overtime (10.44% attrition rate)
- **Attrition based on tenure:** attrition is front-loaded, with an attrition rate of 29.82% in an employee's first 0–2 years, with a steady decline when moving towards more tenured groups, with an attrition rate of 10.38% for 10+ years tenure group.
- **risk of attrition rises consistently with frequency of business travel**, as the attrition rate of frequent travelers sits at 24.91% with 14.96% for rare travelers and 8.00% for non-travelers.

Next Steps

- Retention of sales representatives should be the highest priority, explore career path options to discuss with new employees and sales representatives, and gather more data on sales representatives who depart, through exit interviews or similar means.
- check the distribution of workload in teams which contribute a lot of overtime hours, and check the possibility of balancing workload with employees who do not contribute overtime hours.
- assess the travel demands of employees who frequently travel and check opportunities for reducing the need to travel for frequent travelers.
- Data can be explored further to produce predictive models, fields such as Job Satisfaction, education level, and work-life balance can potentially be of great use being used as predictors for predictive machine learning models.
- The inclusion of a date of attrition and date of join fields can allow for date-based trend analysis, identifying seasonality in attrition, if it exists, and correlating events with spikes in attrition.